

Interchange Exposure, and the report service

Where the build stands after today, what a member can call, and the nine things that need a decision or an action from you before this goes in front of one.

PREPARED BY	Claude, for Faith Birgen
DATE	16 September 2026
STATUS	Build in progress
REGISTRY	17 members · 14 consents
AUDIT	43 calls · 35 log entries
FILTERS	4 published

MEMBERS REGISTERED 17 17 with a signing key	BOOKS PUBLISHED 4 3002, 3005, AXE-3003, AXE-3004	BORROWERS CONTRIBUTED 75,025 tokenised — no identifier reached the Registry	LIVE EXPOSURE HELD KES 493M across contributing members	LOG ENTRIES 35 hash-chained, re-verifiable
--	---	--	--	---

WHAT CHANGED TODAY

The Registry was rebuilt on a new database and reseeded, the founding cohort and both Micromart and Axe entities were registered with their real signing keys, the full report engine was built and proven against a real Metropol file, and the first member books were published from live Serviceconnect over Tailscale. Thirteen of Metropol's fourteen report types were confirmed entitled on Micromart's production contract, free of charge, using a probe identity that buys nothing.

The network as it stands

Every member below is registered in the Registry with an Ed25519 public key. A member cannot call anything without one.

MEMBER	CODE	STATUS	BOOK PUBLISHED	BORROWERS	EXPOSURE HELD
Micromart Africa	KE/LENDER/3002	ACTIVE	2026-09-16 00:07	59,968	KES 355,114,906
Micromart Fintech	KE/LENDER/3005	ACTIVE	2026-09-15 23:57	153	KES 1,380,910
Axe - Boresha	KE/LENDER/AXE-3003	SHADOW	2026-09-16 00:12	14,792	KES 128,080,380
Axe - Stawi	KE/LENDER/AXE-3004	SHADOW	2026-09-16 00:12	112	KES 8,568,855

Axe is in the shadow period by design: contributing but not yet querying. Promotion is a governance action, recorded with who decided it and why — not a config change.

Exposure, measured on the real books

Every figure below was produced today against live Serviceconnect data, not a fixture.

POSITIONS PUBLISHED 75,025 tokenised borrowers with open money	HELD BY 2+ MEMBERS 2,252 the borrowers the product exists for	OVERLAP RATE 3.0% of all published positions	NODE TIMEOUTS 0 no member failed to answer	P95 1,182 ms outside the 400 ms target — see action 1
---	--	---	---	--

READ THE LATENCY HONESTLY

At the blueprint's 250 ms per-member budget, every node timed out and the query returned in 290 ms having found nothing. That is fast and worthless, and the broker correctly reported it as partial rather than as 'no exposure' — which is the single most dangerous answer this service could give.

Raising the per-member budget produced real answers: 25 of 25 sampled borrowers were confirmed at two lenders, with no timeouts and no errors. The 1,182 ms is therefore a measurement of DISTANCE — each node read is a round trip to a Postgres in Ireland — not of the design. The fan-out itself, and the Bloom screening, take about a millisecond.

What a member can call today

One signed endpoint, three formats. Ecosystem reports are free to a contributing member; bureau-backed reports are the contract holder's cost plus a stated fee.

TYPE	REPORT	SOURCE	PER CALL	STATE
1	Identity Verification Is this ID real, and is it the person in front of me?	Metropol	KES 12	live
2	Delinquency Status Are they in default right now, anywhere in this ecosystem?	Member books	free	live
3	Interchange Score What is the network's number on this person, and why?	Both	KES 30	specified
11	Cashflow & Affordability What can they actually afford to repay each month?	Both	KES 75	specified
12	Credit File Everything on file, read as a decision rather than as a dump.	Both	KES 250	live
20	Ecosystem Exposure Is this borrower, who looks clean to me, servicing loans elsewhere right now?	Member books	free	live
21	Intent Signal Who is this borrower shopping with right now?	Member books	free	specified
22	Contactability When and how do we actually reach this person?	Member books	free	specified
23	Cohort Benchmark How does this borrower compare with their peers across the ecosystem?	Member books	free	specified

Bureau costs are indicative until Metropol's tariff sheet is in the vault — see the action list.

METROPOL ENTITLEMENT, SWEPT LIVE ON 16 SEP 2026

no report was bought

13

Entitled 13

Refused (report 22, E029) 1

The probe uses a dummy identity that exists on no production file, so an entitled subscription answers 'identity not found' and an unentitled one answers 'unauthorized report'. Entitlement is therefore measurable for free, and can be re-run whenever a contract changes.

What is needed from you

Ordered by what blocks a live demo first. Everything else in this document is done and running.

1. DECIDE WHERE THE REGISTRY RUNS, AND PUT ITS DATABASE NEXT TO IT

WHY — The p95-under-400ms target cannot be met from here. The Registry's Postgres is in Ireland and a single gate call costs about two seconds of round trips from Nairobi. That is a measurement of DISTANCE, not of design — the fan-out itself takes milliseconds. Demoing on this workstation hides it; putting it in front of a member does not.

WHAT TO DO — The blueprint already names the answer: vmi3298281 is on the tailnet, serves 80/443, carries no database and has free capacity. Registry plus its Postgres on that host, and the number becomes real.

2. TURN ON BUREAU SELLING, AND CONFIRM THE DAILY CEILING

WHY — The report service currently replays a captured Metropol answer so that nothing is billed while it is being built. The live path is written, signed and typechecked, but switched off — deliberately, because it spends Micromart's money.

WHAT TO DO — In connected-suite/.env uncomment INTERCHANGE_BUREAU_SELL=1.

INTERCHANGE_BUREAU_DAILY_CAP is 50 pulls per 24 hours; raise or lower it before the demo. Then remove INTERCHANGE_BUREAU_REPLAY_DIR from the console's .env and the same call goes to the real bureau.

3. GET METROPOL'S TARIFF SHEET INTO THE VAULT

WHY — Every bureau price in the catalogue is a placeholder carrying the market's shape. A member will ask what a credit file costs in the first meeting, and 'indicative' is a weak answer when the rest of the product is this precise.

WHAT TO DO — Ask Ambale for the commercial rate card, then enter it once as CrbConfig.tariff. Every projection, quote and invoice line re-prices itself — no code change.

4. DECIDE WHETHER AXE KNOWS, AND WHETHER AXE MAY QUERY

WHY — Axe's two books were read live today and can be published in minutes. Axe is registered in SHADOW, which is correct — contributing without querying is the reciprocity apprenticeship. But a competitor's book is being read, and whether they have agreed to that is a question the software cannot answer.

WHAT TO DO — If they have agreed, promote them with the CLI, which records the decision, who made it and why. If they have not, the demo runs on Micromart's two entities alone and still shows cross-member exposure.

5. BRING THE SQL RELAY NODE BACK, OR ACCEPT DIRECT TDS FOR THE DEMO

WHY — servicesuite-in has been offline for nine hours, so the ingest ran over a direct Tailscale TDS connection from this workstation. That works here and cannot work from Vercel, where the tailnet address has no route.

WHAT TO DO — Wake the node, or plan the demo from a host on the tailnet. If the Registry moves to vmi3298281 (action 1), it is on the tailnet and this stops being a question.

6. CHOOSE THE DEMO BORROWER

WHY — The moment that sells this is one real person who is clean at one lender and three loans deep across the ecosystem. It has to be a real borrower in both books, and it has to be picked deliberately rather than found on stage.

WHAT TO DO — Once both Axe books are published, the overlap can be computed from the published tokens without anybody seeing an identifier. Say the word and I will produce the candidate list.

7. SETTLE CONSENT FOR REAL BORROWERS BEFORE A MEMBER QUERIES IN ANGER

WHY — Every query in the demo mints consent through the console, which is honest for a demo and not a lawful basis at scale. Counsel's condition was that consent is provable for every borrower at every hop.

WHAT TO DO — Wire the Consent Center into the PWA onboarding (the native path) and give members the drop-in component for the bridged path. The wording and the ledger already exist; what is missing is the screen.

8. DECIDE CUSTODY OF THE ECOSYSTEM OPRF KEY

WHY — It sits in an .env file. Rotating it re-tokenises every borrower in the ecosystem, so it is effectively unrotatable once real members depend on it — which they now do, because 60,000 tokens were derived under it today.

WHAT TO DO — A KMS or an HSM before the second member joins. This is cheap now and very expensive later.

9. CLOSE THE REGISTRY'S OWN ROW-LEVEL SECURITY GAP

WHY — The Registry connects as postgres, which carries BYPASSRLS — the same fault already measured in the LMS. Tenant scoping is app-level only, so a query that forgets its member scope returns everything rather than nothing.

WHAT TO DO — Create an interchange_app role with NOBYPASSRLS, grant it with the bootstrap script's --grant flag, and move DATABASE_URL onto it. The pattern is already written for the LMS.

What is not true yet

Stated plainly, because the value of this document is that it can be trusted.

EVIDENCE BLOCKS		what this file is assembled from
● Signed, consented, logged member-to-member calls	PRESENT	19 acceptance checks, all passing against the live Registry.
● Report engine — JSON, HTML and PDF, gated and logged	PRESENT	17 acceptance checks, including four attacks, all passing.
● Metropol entitlement — 13 of 14 report types	PRESENT	Swept live today at no cost.
○ Bureau reports from a live pull	UNAVAILABLE	Written and switched off. Action 2 turns it on.
● Exposure across members on real books	PRESENT	All four books published — 75,025 positions — and 2,252 borrowers confirmed at more than one member. 25 of 25 sampled overlaps answered with no timeouts and no errors.
○ p95 under 400ms	UNAVAILABLE	1,182 ms measured, with zero node timeouts. The budget is missed on distance to the database, not on the fan-out, which takes about a millisecond — see action 1.
○ RFC 3161 timestamping of the message log	UNAVAILABLE	The chain proves order and integrity, not wall-clock time.
○ Consent captured from borrowers themselves	UNAVAILABLE	Minted by the console today. Action 7.
○ Interchange Score, Intent, Contactability, Cohort	UNAVAILABLE	Specified in the catalogue, not yet answering.

THE ONE-SENTENCE PITCH THIS NOW SUPPORTS

A lender asks one signed question and learns, in under half a second and free at the point of use, whether a borrower who looks clean to them is three loans deep across the ecosystem — and can buy the bureau file in the same call, in the same envelope, without holding a bureau contract of their own.